

# Identification of Novel Cryptic Pockets in RAC1

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## Abstract

The work described here attempts to locate cryptic pockets in the RAC1 protein.

More to be written

## Introduction

- importance of GTPases, their switch mechanism and their interactions with GEFs, GAPs and GDIs
- structural biology information about RAC1 and protein-protein interactions with GEFs, GAPs and GDIs
- Therapeutic interventions on RAC1 and their disadvantages
- Binding site information and changes in the binding sites, if any are reported
- Are there any reports on cryptic pockets for RAC1

## Methods

- List of PDBS used

- Protocol for AMBER MD simulations
- Protocol for POVME2
  - pocket identification
  - pocket volume measurements
- fragment placement and trying to keep the pocket open to target drug-like molecules
  - Type of pocket and the type of fragment more suitable to keep it open
- Analysis of other MD observations to support pocket volume changes
  - RMSDs for switch region 1 and 2
  - (perhaps) PCA analysis to explain the switch region dynamics and the pocket volumes

## Results and discussion

- Try to look for a connection/relation between the following
  - switch region dynamics with different binding partners
  - switch region dynamics with pocket volumes
  - If we find different pockets that might open and close with different binders

## Outline

The document layout should follow the style of the journal concerned. Where appropriate, sections and subsections should be added in the normal way. If the class options are set correctly, warnings will be given if these should not be present.

## References

The class makes various changes to the way that references are handled. The class loads `natbib`, and also the appropriate bibliography style. References can be made using the normal method; the citation should be placed before any punctuation, as the class will move it if using a superscript citation style.<sup>?</sup> The use of `natbib` allows the use of the various citation commands of that package: `?` have shown something, in `?`, or as given by Ref. `?`. Long lists of authors will be automatically truncated in most article formats, but not in supplementary information or reviews.<sup>?</sup> If you encounter problems with the citation macros, please check that your copy of `natbib` is up to date. The demonstration database file `achemso-demo.bib` shows how to complete entries correctly. Notice that “et al.” is auto-formatted using the `\latin` command.

Multiple citations to be combined into a list can be given as a single citation. This uses the `mciteplus` package.<sup>?</sup> Citations other than the first of the list should be indicated with a star. If the `mciteplus` package is not installed, the standard bibliography tools will still work but starred references will be ignored. Individual references can be referred to using `\mciteSubRef`: “ref. `??`”.

The class also handles notes to be added to the bibliography. These should be given in place in the document.<sup>?</sup> As with citations, the text should be placed before punctuation. A note is also generated if a citation has an optional note. This assumes that the whole work has already been cited: odd numbering will result if this is not the case.<sup>?</sup>

## Floats

New float types are automatically set up by the class file. The means graphics are included as follows (Scheme `??`). As illustrated, the float is “here” if possible.

Charts, figures and schemes do not necessarily have to be labelled or captioned. However, tables should always have a title. It is possible to include a number and label for a graphic without any title, using an empty argument to the `\caption` macro.

Your scheme graphic would go here: `.eps` format  
for  $\text{\LaTeX}$  or `.pdf` (or `.png`) for pdf $\text{\LaTeX}$   
CHEMDRAW files are best saved as `.eps` files:  
these can be scaled without loss of quality, and can be  
converted to `.pdf` files easily using `eps2pdf`.

Scheme 1: An example scheme

As well as the standard float types `table`  
and `figure`, the class also recognises  
`scheme`, `chart` and `graph`.

Figure 1: An example figure

The use of the different floating environments is not required, but it is intended to make document preparation easier for authors. In general, you should place your graphics where they make logical sense; the production process will move them if needed.

## Math(s)

The `achemso` class does not load any particular additional support for mathematics. If packages such as `amsmath` are required, they should be loaded in the preamble. However, the basic  $\text{\LaTeX}$  `math(s)` input should work correctly without this. Some inline material  $y = mx + c$  or  $1 + 1 = 2$  followed by some display.

$$A = \pi r^2$$

It is possible to label equations in the usual way (Eq. ??).

$$\frac{d}{dx} r^2 = 2r \tag{1}$$

This can also be used to have equations containing graphical content. To align the equation

number with the middle of the graphic, rather than the bottom, a minipage may be used.

As illustrated here, the width of  
the minipage needs to allow some  
space for the number to fit in to. (2)

## Experimental

The usual experimental details should appear here. This could include a table, which can be referenced as Table ??.

Table 1: An example table

| Header one  | Header two  |
|-------------|-------------|
| Entry one   | Entry two   |
| Entry three | Entry four  |
| Entry five  | Entry five  |
| Entry seven | Entry eight |

Adding notes to tables can be complicated. Perhaps the easiest method is to generate these using the basic `\textsuperscript` and `\emph` macros, as illustrated (Table ??).

Table 2: A table with notes

| Header one               | Header two |
|--------------------------|------------|
| Entry one <sup>a</sup>   | Entry two  |
| Entry three <sup>b</sup> | Entry four |

<sup>a</sup> Some text; <sup>b</sup> Some more text.

The example file also loads the optional `mhchem` package, so that formulas are easy to input: `\ce{H2SO4}` gives H<sub>2</sub>SO<sub>4</sub>. See the use in the bibliography file (when using titles in the references section).

The use of new commands should be limited to simple things which will not interfere with the production process. For example, `\mycommand` has been defined in this example, to give italic, mono-spaced text: *some text*.

## Extra information when writing JACS Communications

When producing communications for *J. Am. Chem. Soc.*, the class will automatically lay the text out in the style of the journal. This gives a guide to the length of text that can be accommodated in such a publication. There are some points to bear in mind when preparing a JACS Communication in this way. The layout produced here is a *model* for the published result, and the outcome should be taken as a *guide* to the final length. The spacing and sizing of graphical content is an area where there is some flexibility in the process. You should not worry about the space before and after graphics, which is set to give a guide to the published size. This is very dependant on the final published layout.

You should be able to use the same source to produce a JACS Communication and a normal article. For example, this demonstration file will work with both `type=article` and `type=communication`. Sections and any abstract are automatically ignored, although you will get warnings to this effect.

## Acknowledgement

Please use “The authors thank ...” rather than “The authors would like to thank ...”.

The author thanks Mats Dahlgren for version one of `achemso`, and Donald Arseneau for the code taken from `cite` to move citations after punctuation. Many users have provided feedback on the class, which is reflected in all of the different demonstrations shown in this document.

## Supporting Information Available

This will usually read something like: “Experimental procedures and characterization data for all new compounds. The class will automatically add a sentence pointing to the information on-line:

## References

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## TOC Graphic

Some journals require a graphical entry for the Table of Contents. This should be laid out “print ready” so that the sizing of the text is correct. Inside the tocentry environment, the font used is Helvetica 8 pt, as required by *Journal of the American Chemical Society*. The surrounding frame is 9 cm by 3.5 cm, which is the maximum permitted for *Journal of the American Chemical Society* graphical table of content entries. The box will not resize if the content is too big: instead it will overflow the edge of the box. This box and the associated title will always be printed on a separate page at the end of the document.